International Dairy Journal 121 (2021) 105138

Contents lists available at ScienceDirect

International Dairy Journal

journal homepage: www.elsevier.com/locate/idairyj

Effect of hurdle technology of gentle pasteurisation and drying process on bioactive proteins, antioxidant activity and microbial quality of cow and buffalo colostrum

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ARTICLE INFO

Article history:
Received 5 March 2021
Received in revised form
5 June 2021
Accepted 7 June 2021
Available online 19 June 2021

ABSTRACT

The effect of various pasteurisation conditions (30–60 min; 57–63 °C) and drying method (spray- and freeze-drying) on bioactive-proteins (immunoglobulin-G and lactoferrin), antioxidant activity, and microbial quality of cow and buffalo colostrum was investigated. Although *increased* pasteurisation *time and temperature* had a significant effect on IgG, the most pronounced reduction was observed in lactoferrin ($P < 0.05$). In cow and buffalo colostrum with increasing pasteurisation temperature from 57 to 63 °C, the DPPH (2,2-diphenyl-1-picrylhydrazyl) radical scavenging activity (DRSA) was decreased by 33% and 37%, respectively. The reduction trend of the ferric-reducing ability of plasma (FRAP), was more outstanding than DRSA. Although freeze-drying was provided a little higher concentration of bioactive proteins, the difference between the two methods was not significant. The combination of gentle-pasteurisation (57 °C, 30 min) and drying improves microbial quality, whereas a reasonable level of bioactive compounds and antioxidant activity were preserved.

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1. Introduction

Human milk is considered as a rich source of bioactive proteins, polyunsaturated fatty acids, oligosaccharides, vitamins, and minerals. These compounds exhibit development, nutritional and immune effects to protect the infant against infections and to assist in the development of the infant's intestinal and immune system (Hassiotou et al., 2013). When mother's milk is low or not available, infant formula (IF) can be prepared based on bovine milk. However, the levels of macronutrients of bovine milk are much lower than human milk and should be manipulated (Al-Mashikhi & Nakai, 1987; Su et al., 2017).

Colostrum is the initial mammary secretion obtained during parturition and the first few (1–4) days of birth (Tripathi & Vashishtha, 2006). It is well known for an elevated amount of nutrients, antibodies, antioxidants and growth factors. Colostrum contains a wide range of bioactive compounds such as immunoglobulins (Igs), lactoferrin (LF), lactoperoxidase (LP), lysozyme, and growth factors (Tripathi & Vashishtha, 2006). Therefore, colostrum has a good potential for use in IF, health beverages, dietary supplements and novel functional foods and drugs (Borad & Singh, 2018). However, various studies show that the microbial load of colostrum samples collected generally exceeds the industry acceptable threshold of 5 log cfu mL^{-1} total plate count (Donahue et al., 2012; McGuirk & Collins, 2004; McMartin et al., 2006). Among all processing technologies, such as freezing, chemical additives, pasteurisation, microwave, dehydration, etc., heat treatment and pasteurisation of fresh colostrum is one of the most common methods to reduce bacterial pathogens (Borad & Singh, 2018).

Pasteurisation of colostrum samples is typically conducted either in low temperature/long time (LTLT) pasteurisation (generally at 63 °C for 30 min) or in standard high temperature/short time (HTST) pasteurisation (at 72 °C for 15 s) (Chatterton et al., 2020; Donahue et al., 2012; McMartin et al., 2006; Nguyen, Currie, Ren, Bering, & Sangild, 2019). Although this process improves the microbial quality of product, it causes undesirable changes in nutritional, functional and physicochemical properties of milk and colostrum samples (Borad & Singh, 2018; Liu, Zhang, Han, Zhang, & Zhou, 2020b; Liu, Zhang, Zhang, Hettinga, & Zhou, 2020c). However, the extent of these changes depends on thermal and

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https://doi.org/10.1016/j.idairyj.2021.105138
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environmental conditions of the product. Heat-treatment (especially temperatures above 60 °C) initiates the unfolding of bioactive proteins and leads to substantial denaturation (Gelsinger, Jones, & Heinrichs, 2015; Xiong, Li, Boeren, Vervoort, & Hettinga, 2020). It was reported mild pasteurisation to enhance the microbial quality of colostrum led to a limited loss of Igs (Liu et al., 2020b; McMartin et al., 2006).

On the other hand, the shelf life of liquid colostrum is limited. Moreover, the majority of supplements and nutraceuticals are commercially available in dried form due to more stability during production, storage and transportation as well as these ingredients and compounds in powder form can be easily added to a wide range of food products and formulations (Borad & Singh, 2018; Chen, Wang, Liu, & Gong, 2017; Yonekura, Sun, Soukoulis, & Fisk, 2014). Colostrum powders are usually prepared by freeze- or spray-drying method. The spray drying (SD) method is considered as a low-cost, fast (5–30 s), simple and easy to scale-up dehydration method which is commonly used to dry various high-value products such as milk, colostrum, yeast, etc., in industry. However, thermal denaturation of bioactive compounds of colostrum, specifically bioactive proteins and peptides is a major challenge in developing spray dried colostrum (Borad & Singh, 2018). Therefore, the development of milder processes is more attractive. The freeze-drying process, as a milder procedure can be considered. Thus, the processing condition (time and temperature of pasteurisation, SD and FD) may affect the bioactivity of various colostrum samples. However, the effect of the different pre-processing techniques remains poorly studied and not verified. It remains to be investigated if pasteurisation at a lower temperature (<60 °C) followed by dehydration could induce less damage to proteins while preserving the capacity to reach acceptable bacterial limits.

The objective of this study was to investigate the effect of various pasteurisation conditions (time and temperature) and drying method (SD and FD) on key bioactive proteins, antioxidant activity and microbial quality of cow and buffalo colostrum samples and to provide the best condition for preserving the bioactive compounds of colostrums.

2. Materials and methods

2.1. Colostrum handling

Fresh cow colostrum was collected from Holstein cows on one livestock in Noor city (Mazandaran, Iran). Buffalo colostrum was obtained from buffalos breeding farm in the Miankaleh region of Mazandaran province (Iran). Colostrum samples were collected during the first 24 h of lactation. The samples were stored at −20 °C and immediately transferred to the Dairy Research Laboratory of Noor Islamic Azad University (Noor, Mazandaran, Iran).

2.2. Pasteurisation

The samples were gently thawed overnight. The pasteurisation of samples was done based on a gentle LTLT process at three different temperatures (57, 60, and 63 °C) and for various holding times (i.e., 30, 45, and 60 min). These times and temperatures were selected based on the literature review. The temperature–time combinations applied in the current study were shown in Table 1. All samples were prepared in triplicate.

2.3. Dehydration

2.3.1. Spray-drying

Spray-drying is the most common technology to produce different dairy powders and other ingredients. The pasteurised samples were immediately spray-dried at feed temperature 54 ± 3 °C and feed rate of 160 mL h^{-1} using a pilot-scale spray dryer (model SD-05, Lab-Plant, Huddersfield, UK). Inlet and air temperatures were maintained at 123 ± 2 and 48 ± 2 °C respectively by drying airflow rate of 0.78 m^3 min^{-1} (Borad & Singh, 2018). The moisture contents of samples were within the acceptable limit (≤ 5%) expected for whole milk powder (CODEX, 2014).

2.3.2. Freeze-drying

Freeze-drying is considered as one of the most preferred methods to dehydrate heat-sensitive materials. The freeze-dried powder was prepared by lab-scale benchtop Freeze-Dryer System (model VUCO5, ZIRBUS, Duesseldorf, Germany). The pasteurised colostrum was subjected to freeze-drying at −50 °C with low pressure (Klobasa, Goel, & Werhahn, 1998). The atmospheric air temperature and relative humidity were kept at 28 ± 2 °C and 52 ± 1%, respectively. The dried powder (moisture content < 5%) was collected and packaged in the dark glass bottle and stored at −20 °C until analysis.

2.4. Chemical analysis

The total protein, fat, total solids (T.S), ash and lactose of colostrum samples were estimated using standard AOAC methods (Horwitz, 2000).

2.5. IgG content

IgG content of colostrum samples was determined based on the enzyme-linked immunosorbent assay (ELISA) technique. They were samples diluted for analysis IgG concentration using *Bovine IgG* ELISA Core Kit, pink-ONE, (KOMA Biotech, Seoul, Korea) according to the procedure prescribed by the manufacture. The absorbance of samples was recorded at 450 nm by a multimode plate reader (Tecan Infinite M200 PRO, Männedorf, Switzerland).

2.6. Lactoferrin content

After defatting the colostrum samples, isolation of lactoferrin from the concentrated proteins was done using CM-Sephadex-C 50 cationic exchange and fast protein liquid chromatography (FPLC) based on a gradient of solvent and the method suggested by Moatsou et al. (2008). The purity of the lactoferrin was investigated by sodium dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE), showed only one band (Sakai, Kobayashi, & Yoshida, 1995). Lactoferrin concentration was determined by the Bradford assay and the absorbance of each sample solution was read at 595 nm (Bradford, 1976).

2.7. DPPH radical scavenging activity (DRSA)

The DPPH [2,2-diphenyl-1-picrylhydrazyl (90.0%; Sigma Chemicals Co, St, Louis, MO, USA)] radical scavenging activity was determined according to the procedure reported by Bersuder, Hole, and Smith (1998) with slight modification as described by Oussaief et al. (2020). For this purpose, 1.0 mL of each colostrum sample at various concentrations (1.0–7.0 mg mL^{-1}) was mixed with 1.0 mL of an ethanol solution of DPPH (125 μM). The mixtures were incubated for 60 min in darkness at room temperature. For the control, distilled water was used rather than the sample. The absorbance of the resultant solution was read at 517 nm using a UV–visible spectrophotometer (UV-6715; Jenway Co., *Staffordshire, UK*).

The DPPH radical-scavenging activity (%) was calculated according to Eq. (1):

$$\text{DPPH scavenging activity}(\%) = \left((A_{Blank}) - A_{Sample} \Big/ A_{Blank}\right) \times 100 \quad (1)$$

where A_{Blank}: the absorbance of DPPH solution without sample; A_{sample}: the absorbance of DPPH solution with extract (Bersuder et al., 1998; Oussaief et al., 2020).

2.8. Ferric reducing ability of plasma (FRAP)

The antioxidant potential of samples was also estimated by ferric reducing ability of plasma (FRAP) assay according to the Benzie and Strain (1996) protocol. The reagent was produced by mixing 10 volumes of acetate buffer (300 mM, pH 3.6), with 1 volume 2,4,6-tripyridyl-s-triazine solution (10 mM) and 1 vol $FeCl_3 \cdot 6H_2O$ (20 mM). Twenty millilitres of colostrum sample were added into 1.5 mL fresh FRAP reagent and incubated at 25 °C for 10 min the change in absorbance was read at 593 nm. Aqueous solutions of ferrous sulphate heptahydrate ($FeSO_4 \cdot 7H_2O$) were used to illustrate the linear calibration curve.

2.9. Microbial analysis

In this study, for monitoring the microbiological quality of cow and buffalo colostrum samples, the total plate count (TPC), Total coliforms count (TCC), *E. coli* count and yeast and mould count as hygiene indicators were enumerated for evaluation of the effects of various treatments on microbiological quality. These analyses were conducted according to ISO:4833-1: 2013 method. The microbial load was recorded in terms of log colony forming units (cfu) per milliliter ($\log_{10}$ cfu mL^{-1}) (ISO, 2013).

2.10. Statistical analysis

All analyses were conducted at least in three replicate measurements and the results were reported as mean values and their standard division (mean ± SD). Statistical analyses were determined ANOVA/General Linear Model (GLM). The significant differences between the means were performed by Tukey's test ($P < 0.05$). The chemical composition of both colostrum samples was analysed using an independent t-student. Statistical calculations were performed in Minitab Version 17 statistical software (Minitab Inc. Pennsylvania, USA).

3. Results and discussion

3.1. Chemical composition of colostrum samples

The comparison between total protein, fat, lactose, ash and total solids concentration of buffalo and cow colostrum samples is presented in Table 1. The results showed that there was no significant difference between the total solid and ash content of the two samples ($P > 0.5$) while fat, total protein and lactose content were significantly higher in buffalo colostrum than cow one ($P < 0.05$). In general, previous studies have reported a relatively wide range of chemical composition among colostrum samples (Borad & Singh, 2018). However, the findings were comparable with previously published results (Abd El-Fattah, Abd Rabo, El-Dieb, & El-Kashef, 2012; Abdel-Ghany & Zaki, 2018; Kim et al., 2011).

3.2. The effect of pasteurisation conditions on IgG content

Immunoglobulins are a group of proteinaceous compounds with protective biological activities. IgG acts as antibodies, improves systemic immunity and provides protection against a wide range of bacterial toxins (Borad & Singh, 2018; McConnell, Buchan, Borissenko, & Brooks, 2001). Table 2 presents the effect of pasteurisation time and temperature on bioactive protein content and antioxidant activity of both colostrums. As shown in this table, by increasing the time and temperature of pasteurisation the IgG concentration was significantly decreased ($P < 0.05$). At a temperature around or above T_m (>60 °C) IgG molecules were unfolded, followed by the irreversible aggregation of the unfolded IgG molecules (Vermeer & Norde, 2000). Therefore, in cow colostrum, around 5.8% and 12.95% IgG loss was observed, when pasteurisation temperature changed from 57 to 60 and 63 °C for 30 min, respectively. In buffalo colostrum, there was a mean loss of 9.27% and 10.27% of IgG. Borad et al. (2021) reported that HTST (72 °C for 15 s) and LTLT (63 °C for 30 min) denatured 56.4 and 24.6% of initial IgG, respectively. However, McMartin et al. (2006) suggested that bovine colostrum could be safely heated to 60 °C for 120 min without significantly decreasing concentration in IgG (around 1.2%). About the effect of pasteurisation time on IgG content of colostrum, Hesami, Shahraki, Zakian, and Ghalamkari (2021) showed that at 60 °C, the colostral IgG level decreased with increasing heat-treatment time by 2.16 ± 0.62, 6.74 ± 0.78, and 11.98 ± 1.03% for 30, 60, and 90 min pasteurisation, respectively. The difference between various results may be due to differences in chemical composition and type of colostrum. Overall, current findings were supported by the previous studies on the effect of pasteurisation conditions on IgG loss of various colostrums (10–40% loss after pasteurisation treatment) (Godden et al., 2003; McMartin et al., 2006; Meylan et al., 1996; Nguyen et al., 2019).

Moreover, the difference between the IgG content of the two colostrums (cow and buffalo) was significant ($P < 0.05$). Although the IgG level of buffalo colostrum was significantly higher than that in cow colostrum, buffalo colostrum was more sensitive than that of cow colostrum to heat treatment. For example, when buffalo samples pasteurisation at 63 °C for 60 min the loss of IgG was above 30% compared with heat treatment at 57 °C and 60 min. But in this same condition, there was around 12% IgG loss in cow colostrum. The susceptibility of IgG isotypes (IgG_1 and IgG_2) to denaturation was not similar. IgG_2 was more sensitive due to the presence of two methionine substitutions that do not exist in IgG_1 (Latypov, Hogan, Lau, Gadgil, & Liu, 2012). The difference between IgG isotypes content of various colostrum may be a reason for different thermal susceptibility of IgG in cow and buffalo colostrum. Moreover, the differences in concentrations of chemical composition and pH of the samples were another reason for this variation. The competition of lactose, minerals, lipids other proteins with IgGs for absorbing water molecules may improve the thermal stability of this bioactive protein (Borad et al., 2021).

Table 1
The chemical composition (%) of buffalo and cow colostrums.[a]

Sample	Fat**	Protein**	Lactose**	Total solids^{NS}	Ash^{NS}	pH**
Buffalo	10.81 ± 0.50	7.97 ± 0.76	3.78 ± 0.06	23.93 ± 1.46	0.97 ± 0.02	6.04 ± 0.02
Cow	6.11 ± 0.78	12.91 ± 3.75	3.21 ± 0.41	23.33 ± 4.11	1.01 ± 0.08	6.26 ± 0.09

[a] Values are the mean ± standard deviation (SD); NS, no significant difference between samples ($P > 0.05$); **, significant difference between samples at level of 1%.

Table 2
The effect of various LTLT pasteurisation conditions on bioactive proteins and antioxidant activity of cow and buffalo colostrum.[a]

Time	57 °C		60 °C		63 °C	
	Buffalo	Cow	Buffalo	Cow	Buffalo	Cow
IgG (mg mL^{-1})						
30 min	37.03 ± 0.67^{a}	31.51 ± 0.83^{cd}	33.60 ± 0.849^{bc}	29.68 ± 0.59^{de}	33.06 ± 0.38^{bc}	27.43 ± 0.7^{de}
45 min	34.97 ± 0.25^{ab}	30.02 ± 1.10^{de}	32.27 ± 0.891^{bcd}	26.77 ± 0.49^{fg}	27.49 ± 0.79^{ef}	22.38 ± 0.54^{hi}
60 min	31.86 ± 1.05^{cd}	25.99 ± 0.44^{fg}	26.66 ± 0.62^{fg}	24.19 ± 0.41^{gh}	21.11 ± 0.83^{i}	22.95 ± 0.63^{hi}
LF (mg mL^{-1})						
30 min	1.075 ± 0.04^{a}	1.07 ± 0.05^{a}	0.74 ± 0.04^{cd}	0.72 ± 0.01^{cd}	0.53 ± 0.03^{efg}	0.45 ± 0.05^{fgh}
45 min	0.92 ± 0.01^{b}	0.96 ± 0.01^{ab}	0.65 ± 0.4^{de}	0.63 ± 0.01^{de}	0.41 ± 0.04^{ghi}	0.32 ± 0.04^{hi}
60 min	0.84 ± 0.14^{bc}	0.83 ± 0.03^{bc}	0.57 ± 0.03^{ef}	0.52 ± 0.04^{efg}	0.36 ± 0.03^{hi}	0.27 ± 0.04^{i}
DRS (%)						
30 min	70.85 ± 0.78^{a}	64.83 ± 1.86^{bc}	59.78 ± 0.64^{cd}	48.57 ± 1.81^{gh}	44.95 ± 1.05^{ghi}	43.26 ± 1.01^{hij}
45 min	66.99 ± 0.57^{ab}	55.28 ± 2.87^{de}	43.93 ± 1.67^{ghi}	46.36 ± 0.19^{gh}	37.95 ± 1.63^{klm}	40.9 ± 1.12^{ijk}
60 min	54.19 ± 1.4^{ef}	49.19 ± 1.44^{fg}	39.88 ± 0.40^{ijk}	37.63 ± 0.74^{jkl}	32.26 ± 1.44^{lm}	33.17 ± 0.97^{lm}
FRAP (μmol L^{-1})						
30 min	1544.4 ± 45.8^{a}	972.79 ± 6.45^{c}	906.9 ± 38.6^{cd}	718.2 ± 16.0^{fg}	752.0 ± 21.8^{ef}	549.1 ± 27.6^{hi}
45 min	1180.7 ± 31.7^{b}	740.9 ± 41.9^{ef}	766.70 ± 38.8^{ef}	667.9 ± 23.1^{fg}	659.3 ± 36.8 ^{fg}	507.0 ± 15.4 ^{hi}
60 min	835.4 ± 20.0^{de}	609.1 ± 14.4^{gh}	697.36 ± 4.34 ^{fg}	485.22 ± 9.15^{ij}	528.6 ± 20.3 ^{hi}	390.4 ± 27.2^{j}

[a] Abbreviations are: IgG, immunoglobulin G; LF, lactoferrin; DRS, DPPH radical scavenging activity; FRAP, ferric reducing antioxidant power. Values are the mean ± standard deviation (SD); different superscript letters for each parameter indicate the significant difference by Tukey's multiple range test ($P < 0.05$).

3.3. The effect of pasteurisation conditions on LF content

Lactoferrin (LF), as an iron-binding glycoprotein, demonstrates several biological activities, such as antimicrobial, antioxidative, anti-inflammatory, anticancer, and immune regulatory properties (Kanyshkova, Buneva, & Nevinsky, 2001; Levay & Viljoen, 1995; Rascón-Cruz et al., 2021).

Buffalo and cow colostrum samples contain around 1.19 and 1.02 mg mL^{-1} LF, respectively (Fig. 1). There was no significant difference between the lactoferrin content of the two species. Table 2 presents the effect of pasteurisation conditions on the LF content of samples. LF concentration decreased significantly by increasing temperature and time ($P < 0.05$). This can be related to the general protein denaturation during thermal processing. There are three forms of LF including (i) ferric iron-depleted (apo)-, (ii) iron-saturated (holo)- and (iii) partially iron-saturated form. Previous researches showed that apo-form lactoferrin was denatured more rapidly than halo one and the denaturation temperature of purified apo-LF and holo-LF is around 70 and 90 °C (Conesa et al., 2008; Wang, Timilsena, Blanch, & Adhikari, 2019). Thus, during pasteurisation especially at 63 °C, most of the denatured lactoferrin were associated with the apo-LF.

Although *increased time and temperature* of pasteurisation had a significant effect on total IgG, the most pronounced reduction was observed in lactoferrin content. Furthermore, previous research reported that due to the fast rate of aggregation the IgGs become locked in aggregates before complete denaturation and the native Igs are protected by physical entrapment. Moreover, denatured immunoglobulins showed a protective effect on native proteins and increased thermal denaturation temperature (Vermeer & Norde, 2000). On the other hand, the maximum Igs stability was at pH 6.0 (Law & Leaver, 2000) which is the natural pH of both colostrums (Table 1). It may be a possible explanation for the relatively good thermal stability of IgG as compared with LF.

Previous research also reported that denaturation of lactoferrin and IgG followed a similar trend, however, LF was more susceptible than IgG isotypes to pasteurisation conditions (Chatterton et al., 2020; Liu et al., 2020b,c). It was reported that heat treatment at 40 °C or 60 °C did not cause significant changes in human milk lactoferrin (Chang et al., 2013). This may be due to differences in the type of samples and their composition.

As seen in Table 2, the levels of LF in cow and buffalo colostrum decreased by 57% and 51% respectively, as temperature increased from 57 °C to 63 °C. Cow colostrum lactoferrin is slightly more sensitive to heat treatment as compared with buffalo colostrum. LF denaturation increased with increasing temperature and protein content as well as shifting the pH to the isoelectric point of LF. Thus, the higher protein content of cow colostrum than buffalo colosrum is a possible reason for more sensitivity of cow samples than

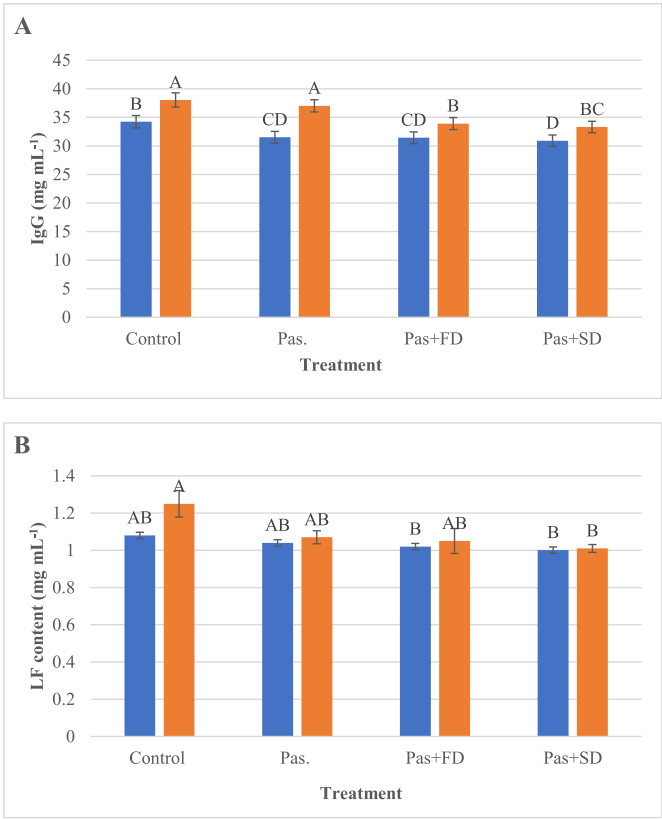

Fig. 1. The effect of drying treatment on (A) immunoglobulin G (IgG) content and (B) lactoferrin (LF) content of cow colostrum (■) and buffalo colostrum (■). Control, fresh colostrum samples; Pas, gentle pasteurisation at 57 °C for 30 min; Pas + FD, combination of gentle pasteurisation and freeze-drying method; Pas + SD, combination of gentle pasteurisation and spray-drying method. The different letters indicate significant differences ($P < 0.05$).

another (Shimo, Xiaoyun, Xuezhi, Lin, & Ping, 2015). Furthermore, the differences between the heat sensitivity of lactoferrin of various colostrum samples may be related to the iron saturation levels of the samples (Liu et al., 2020a). The effect of temperature was more than time on this bioactive protein. For example, when Buffalo colostrum pasteurised at 57 °C for 60 min, its lactoferrin level is still significantly higher than when the pasteurisation was done at 60 °C for 30. Overall, the results confirmed that mild heat treatment retained a reasonable amount of IgG and LF (Fig. 2, Table 2).

3.4. Antioxidant potential of colostrum samples

DPPH assay is based on the electron donation or the acceptance of the hydrogen ion to make stable molecules (Corrêa et al., 2011). DPPH is a suitable marker for screening the scavenging activity related to antioxidant compounds. Furthermore, in the ferric reducing antioxidant power (FRAP) method, the potential antioxidant activity of a compound was estimated based on the ability to reduce ferric ion (Fe^{3+}) to ferrous ion (Fe^{2+}) by donating an electron (Gülçin et al., 2011; Wu & Chang, 2003). Ascorbic acid, α-tocopherol, uric acid, Bilirubin and phenolics are the most important compounds with ferric-reducing activity, but this method may not be suitable for evaluating the antioxidant activity of proteins, glutathione and lipoic acid (Benzie & Szeto, 1999). Since the activities of antioxidants vary in various biological systems; it is necessary to use several methods for evaluating the antioxidant capacity of samples. Therefore, antioxidant activity of cow and buffalo colostrum samples at various heat treatment conditions were assessed based on free radical scavenging activity (DPPH methods) and total antioxidant activity (FRAP assay) methods.

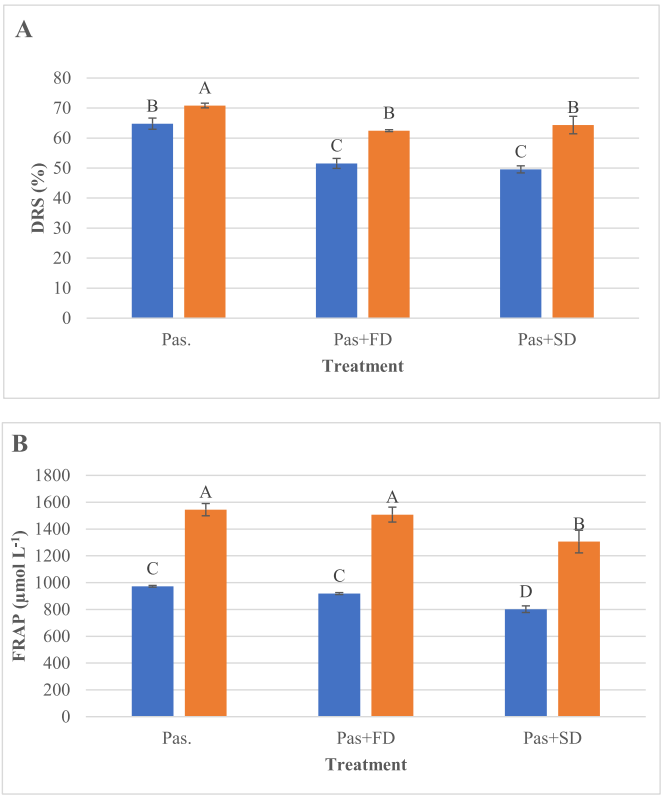

Fig. 2. The effect of drying treatment on (A) DPPH radical scavenging activity (DRS; %) and (B) ferric reducing antioxidant power (FRAP; μmol L^{-1}) of cow colostrum (■) and buffalo colostrum (■). Pas, gentle pasteurisation at 57 °C for 30 min; Pas + FD, combination of gentle pasteurisation and freeze-drying method; Pas + SD, combination of gentle pasteurisation and spray-drying method. The different letters indicate significant differences ($P < 0.05$).

3.4.1. DPPH radical scavenging activity (DRS)

The effect of various LTLT pasteurisation conditions on DRS levels of cow and buffalo colostrum samples is shown in Table 2. The DRS level of samples was significantly decreased by increasing the time and temperature of heat treatment ($P < 0.05$). Moreover, the difference between the two samples was significant at 57 °C and 60 °C for 30 min heat treatment. However, in buffalo colostrum, the decrease rate of the DRS level increased by increasing the time and temperature of pasteurisation, so that the difference between two colostrum samples becomes insignificant gradually (Table 2). In cow and buffalo colostrum with increasing temperature from 57 to 63 °C, the DRS levels decrease 33% (from 64.83 to 43.26%) and 37% from (70.85–44.95%), respectively (pasteurisation time: 30 min). Milk and colostrum proteins are considered as one of the best sources of bioactive peptides with antioxidative, antimicrobial, antithrombotic, anti-inflammatory and immunomodulatory activities (Korhonen, 2010). Several antioxidants including enzymatic (superoxide dismutase, glutathione peroxidase, catalase), non-enzymatic proteins and peptides [lactoperoxidase (LPO), lactoferrin (LF) and ceruloplasmin (CP), whey protein, casein hydrolysate], phenolic compounds and vitamins (A and E) were found in colostrum (Albera & Kankofer, 2009; Przybylska, Albera, & Kankofer, 2007). In this regard, Brijesha and Aparna (2017) reported that depending on peptide concentration, buffalo colostrum showed 58–88% ABTS free radical scavenging (Brijesha & Aparna, 2017). Buffalo peptides were more effective antioxidants than cow colostrum peptides.

During heat treatment, these bioactive compounds may be denatured and lose much of their bioactivity. Moreover, the heat denaturation temperature of bioactive proteins and peptides is usually between 60 and 90 °C (Korhonen, Pihlanto-Leppäla, Rantamäki, & Tupasela, 1998; Silvestre et al., 2008). On the other hand, at temperatures of 60 °C and above, colostrum was coagulated and aggregated. The β-Lg contains a free thiol, which reacting with κ-caseins on the micelle surface during thermal treatments (≥60 °C). On the other hand, α-La and bioactive proteins (e.g., IgG, LF and lactoperoxidase) also can bind with β-Lg and form an aggregate (Nguyen, Sangild, Li, Bering, & Chatterton, 2016). These phenomenons probably reduced the functional activity of the bioactive proteins and antioxidant power of colostrums. The previous investigations indicated that the free peptides induced more bioactivity than interlinked ones (Ashok, Vivek, & Aparna, 2019). Thus, the antioxidant activity of both colostrums was significantly reduced by increasing the time and temperature of pasteurisation.

3.4.2. Ferric reducing ability of plasma (FRAP)

The total antioxidant activity of colostrum samples was determined by FRAP assay. The average reduction of FRAP value by increasing temperature from 57 to 63 °C was found to be 44 and 51%, respectively (pasteurisation time, 30 min; Table 2). Similar findings were also reported on losses of bioactive components and reduce FRAP index in various colostrum samples during heat treatment (Kumar et al., 2015; Tu, Xu, Ren, & Zhang, 2020). The reduction trend of FRAP, in response to increasing temperature and time, was more pronounced than DRS. This difference can be attributed to the fact that tocopherols and phenolic compounds have an outstanding ferric-reducing activity (Benzie & Szeto, 1999) which are more sensitive than several other antioxidant compounds to food processing (Khoshnoudi-Nia, Sharif, & Jafari, 2020).

Overall, although many studies were introduced 60 and especially 63 °C (Ashok et al., 2019; Bartkiene et al., 2020; Chatterton et al., 2020; Godden et al, 2003, 2006; Nguyen et al., 2019;

Trujillo et al., 2007) for the pasteurisation process of colostrums, in practice this temperature greatly reduces the antioxidant potential and bioactive proteins (especially lactoferrin) of colostrum samples. On the other hand, the shelf-life of liquid colostrum was lower than powder one. However, the previous research and our preliminary studies showed that the freeze-drying or spray-drying process alone is not sufficient to ensure acceptable microbial quality of the colostrum samples (Borad, Singh, Meena, & Raghu, 2020; Sotudeh, Rabbani Khorasgani, Etemadifar, & Zarkesh-Esfahani, 2018). Therefore, the use of hurdle technology of gentle pasteurisation and drying method may provide better microbial quality and shelf-life for various colostrum samples. Therefore, in the following sections, the effect of the drying methods after gentle pasteurisation was investigated.

3.5. The effect of drying method on IgG and LF content

Despite spray-drying advantages, because of the possibility of heat-induced damage of bioactive compounds during this process the use of this technique has been challenged for preparing colostrum powder. In this regard, the freeze-drying method may be a good solution to produce colostrum powder with excellent bioactive compound stability. However, high production costs, time-consuming process and difficulties in scaling up restrict the further industrial development of the freeze-drying method (Borad & Singh, 2018; Khoshnoudi-Nia et al., 2020). Thus, the effect of these drying methods on bioactive proteins of colostrum were compared in the current study. As seen in Fig. 1, both drying processes induced only slight protein denaturation and the difference between the two methods in retaining bioactive proteins (IgG and LF) was not significant ($P > 0.05$). Although freeze-drying provided a little higher concentration of the bioactive proteins, spray drying is also a promising and simple technique to retain these valuable compounds. Moreover, the stability of LF against the drying process was more than the IgG. Previous studies were reported that the spray-drying did not affect the LF content of bovine colostrum as compared with IgG (Chatterton et al., 2020). Chelack, Morley, and Haines (1993) showed that 94 and 99% Ig was retained upon spray- and freeze-drying, respectively. Martysiak-Żurowska, Rożek, and Puta (2020) also expressed that although freeze-drying caused a decrease in LF content (about 11.5%), the difference between the LF content of liquid and freeze-dried human milk was not significant. The results also showed that pasteurisation at 57 °C for 30 min had no significant effect on the bioactive protein content. At the same time, pasteurisation can be considered as a pre-heated treatment to facilitate moisture evaporation during the spray-drying process. The above feed temperatures caused gelation and coagulation of colostrum at the nozzle, leading to its blockage. The previous investigation showed that spray-drying and freeze-drying had no major effects on Igs and LF contents (Bartkiene et al., 2020). Overall, the LF and IgG content of samples treated by a combination of gentle pasteurisation and drying was higher than samples heat-treated at 60 °C and above (Table 2; Fig. 1).

3.6. The effect of drying method on antioxidant activity

Fig. 2A shows the effect of drying methods on the DRS values of cow and buffalo colostrums. In the present study, the average scavenging activity of pasteurised cow and buffalo were 64.8 and 70.85%, whereas the average DSR values of freeze-dried samples were 51.56 and 62.47%, respectively ($P < 0.05$). This decrease might be due to the loss of some antioxidant compounds during the process. Similar findings were reported on decreasing DRS value of colostrum during the freeze-drying process (Sharma, Mondal, Raguvaran, Das, & Yadav, 2020). Martysiak-Żurowska, Puta, Rodzik, and Malinowska-Pańczyk (2017) reported that the significant decrease in the total antioxidant capacity during freeze-drying refer to the change in the protective activity of antioxidants rather than the direct destruction of antioxidant. Spray drying also significantly reduced the DRS values of samples ($P < 0.05$). Although there were no significant differences among DRS values of colostrum treated by freeze- and spray-drying, the freeze-drying process had a lower reducing effect as compared with the spray-drying one.

Fig. 2B shows the effect of colostrum type and drying method on the FRAP index. Freeze-drying had no significant effect on the antioxidant activity of colostrum. Previous research showed that oxidation of bioactive proteins and peptides may take place on various acid amine residues, among them methionine residues showed the most sensitivity (Sharma & Graham, 2010; Stadtman, Moskovitz, & Levine, 2003; Suprun et al., 2016). Methionine oxidation (decreasing free methionine content) may be a suitable marker for investigating the level of oxidation occurring during processing (Chatterton et al., 2020). The previous researches showed that the level of free methionine of freeze-dried bovine samples was lower than spray-dried ones (Bartkiene et al., 2020; Xue et al., 2015). Moreover, lactoferrin, as an Iron-related element, is not affected by the freeze-drying process (Gallo et al., 2020).

As seen in Fig. 2B spray drying significantly reduced the FRAP index of samples. Spray drying is detrimental to phenolic compounds and bioactive peptides due to their susceptibility to high temperatures (Dela et al., 2003; Khoshnoudi-Nia et al., 2020).

3.7. The effect of drying method on microbial quality

TPC and TCC are standard methods to investigate the microbial quality of dairy products. The acceptable limit for TPC and TCC of Fresh colostrum is considered 100,000 cfu mL^{-1} (5 log cfu mL^{-1}) and less than 10,000 cfu mL^{-1} (4 log cfu mL^{-1}), respectively (McGuirk & Collins, 2004). However, in the heat-treated colostrum, these thresholds are 20,000 cfu mL^{-1} (4.3 log CFU mL^{-1}) and 100 cfu mL^{-1} (2 log cfu mL^{-1}) for TPC and TCC, respectively (Godden, Lombard, & Woolums, 2019). Table 3 showed the effect of pasteurisation and drying process on microbial quality of both colostrum samples. As shown in this table, the TPC values of fresh colostrum were higher than the acceptable limit of 5 log cfu mL^{-1} and other investigated microbial parameters in the range 2.23–2.87 log cfu mL^{-1}. The difference between the two colostrum samples was not significant at a 5% level ($P > 0.05$). Colostral TPC decreased significantly ($P < 0.05$) up to 2 log cfu mL^{-1} when batches were pasteurised at 57 °C for 30 min. Moreover, the TCC, *E. coli*, yeast, and mould counts were lower than 1 log cfu mL^{-1} after 30 min gentle pasteurisation. Therefore, gentle pasteurisation reached the microbial load of the samples to an acceptable level. In this regard, previous authors also indicated that *E. coli* and Salmonella enteritidis were not detected in the colostrum samples pasteurised at 60 °C for 30 min (Godden et al., 2006; Hesami et al., 2021). However, Sotudeh et al. (2018) determined that heat treatment at 55 °C for 60 min or the use of freeze-drying and spray-drying process alone was not able to provide the acceptable microbial load (TPC < 5 log cfu mL^{-1}) (Sotudeh et al., 2018). Borad et al. (2020) also reported that the total plate count of colostrum samples decreased only 0.5 log cfu mL^{-1} after spray-drying (Borad et al., 2020). Hayes, Hughes, and Greene (2012) detected *Bacillus* species in colostrum powder. However, current results showed that the combination of gentle pasteurisation and drying method had significant effects on the reduction of all studied microbial groups at least one log cfu mL^{-1}, whereas a reasonable level of bioactive compounds was preserved.

Table 3
The effect of gentle pasteurisation combined with drying method on microbial quality of colostrum samples.[a]

Type of treatment	TPC (log cfu mL^{-1})		TCC (log cfu mL^{-1})		*E.coli* (log cfu mL^{-1})		Y&M (log cfu mL^{-1})	
	Cow	Buffalo	Cow	Buffalo	Cow	Buffalo	Cow	Buffalo
Control	5.02 ± 0.54^{a}	5.29 ± 0.41^{a}	2.87 ± 0.42	2.45 ± 0.28	2.23 ± 0.23	2.69 ± 0.52	2.70 ± 0.48	2.61 ± 0.25
Pasteurisation	2.63 ± 0.42^{b}	3.40 ± 0.54^{b}	<1	<1	<1	<1	<1	<1
Freeze-drying	1.19 ± 0.30^{c}	1.77 ± 0.38^{c}	<1	<1	<1	<1	<1	<1
Spray-drying	0.92 ± 0.19^{c}	1.40 ± 0.33^{c}	<1	<1	<1	<1	<1	<1

[a] Abbreviations are: TPC, total plate count; TCC, total coliform count, Y&M, yeast and mould count. Values are the mean ± standard deviation (SD). The different letter superscripts for each parameter indicate the significant difference by Tukey's multiple range test ($P < 0.05$). The difference between two colostrum samples was not significant ($P > 0.05$). Each of freeze and spray drying process was conducted after gentle pasteurisation (at 57 °C for 30 min).

4. Conclusion

The results showed that gentle pasteurisation (at 57 °C for 30 min) had a negligible or slightly negative impact on the bioactive proteins and antioxidant activities of colostrum samples. Considering the obtained results, it seems that the combination of gentle pasteurisation (at 57 °C for 30 min) and freeze-drying or spray-drying methods, as a good hurdle method, could be more effective than pasteurisation at higher time and temperature to maintain bioactivity and to improve microbial quality and shelf life of cow and buffalo colostrum. Moreover, the comparison of two colostrum samples indicated that the bioactivity of buffalo colostrum was more than cow ones.

Declaration of competing interest

The authors declare that no conflict of interest to disclose.

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